Robust Inference in the Presence of Weak Instruments*

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Abstract

Despite a rich literature and well-established results detailing large potential distortions in the presence of weak instruments, practitioners continue to rely on standard t-ratio based inference after insufficiently pre-testing for instrument weakness. Building on Lee et al. (2021), we apply the so-called tF critical value function to adjust standard errors in just-identified, single endogenous variable IV specifications and thereby allow for robust t-ratio based inference. We perform this reassessment in a retrospective manner, for a sample of papers published in American Economic Journals over the past decade and provide insights into the magnitude of adjustments and the changes in significance levels found in this revision.

Keywords: instrumental variables, weak instruments, robust inference, standard error adjustment

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1 Introduction

In many settings, especially when randomized controlled experiments are not possible, researchers must employ alternative methods to estimate causal relationships. One such approach is instrumental variables (IV) estimation. When explanatory variables suffer from endogeneity – a violation of the zero conditional mean of the error term assumption – standard ordinary least squares (OLS) produces biased estimates. As a result, researchers may employ an *instrumental variable* that induces changes in the endogenous explanatory variable but has no other, independent effect on the dependent variable (conditional on a vector of exogenous variables used as controls) to recover an unbiased estimate of the effect of the explanatory variable on the dependent variable. This is possible by using only the variation in the explanatory variable that is uncorrelated with the error term via a decomposition of the explanatory variable using the instrumental variable. Formally, for dependent variable Y, endogenous regressor X_1 and additional explanatory variables X we want to estimate:

$$Y = \beta_1 X_1 + \beta X + u \tag{1}$$

using instrument Z where $cov(Z, u) = 0$ and $cov(Z, X_1) \neq 0$. Then we can recover an unbiased estimate β_1 using two-stage least squares (2SLS) by first regressing X_1 on Z: $X_1 = \pi Z + v$.

Still, the IV estimator is not necessarily unbiased. One of the main challenges in IV is that of "weak" instruments. When the correlation between the instrumental variable and endogenous explanatory variable (conditional on the controls) is close to zero, the instrument is called weak. Under these conditions, 2SLS IV estimation is generally unreliable (Andrews et al., 2019). Bound et al. (1995) pointed out that weak instruments lead to biased estimates in small samples and to a magnification of inconsistency from small violations of the exclusion restriction (i.e. slight correlation of the instrument Z with the initial error term u).

As both the endogenous explanatory variable X and the instrument Z are observable, weak instruments can be directly tested for. Indeed to avoid the aforementioned issues, many researchers "pre-test" for instrument weakness by running an F-test (the null hypothesis being that the instrument is *irrelevant* – or not correlated with the explanatory variable) on the first stage estimates (i.e. $\hat{\pi}$). It is commonplace today for researchers to then rule out weak instrument-based issues in inference if the F-statistic is greater than 10 – though we will come back to this practice in greater detail later on.

Nonetheless, a growing literature in theoretical econometrics, inspired by the canonical proposition by Anderson and Rubin (1949) is producing relevant results to avoid finite-sample biasedness in IV estimation when the instruments are weak. Building on Stock and Yogo (2005) and Lee et al. (2021) for just-identified (one endogenous regressor, one instrument) specifications, this paper analyzes a sample of all publications in the American Economic Journal during the 2015-2019 period (to be extended, time permitting) using instrumental variables, and reassesses their identification framework applying the so-called tF critical value function Lee et al. (2021) for valid inference.

The paper is structured as follows: Section 2 explores the relevant literature on robust inference in the presence of weak instruments further, Section 3 documents the data-gathering process and describes the sample, Section 4 presents the results and methodology used to correct standard errors and confidence intervals, and includes a correlation analysis, and Section 5 concludes.

2 Weak Instruments and Robust Causal Inference

Previous literature has addressed the issue of weak instruments in the accuracy of inference, mainly focusing on multiple instrument models. Anderson and Rubin (1949) first proposed the Anderson-Rubin (AR) bounds which they showed minimized Type II errors in different test settings. Notably, this result has been extended from the baseline Data Generating Process (DGP) with homoskedasticity (Moreira and Poi, 2003; Moreira, 2009) to the generalized cases of heteroskedastic, autocorrelated and/or clustered standard errors (Moreira and Moreira, 2019).

The importance of pre-testing for weak instruments has been largely settled in the last 20 years. Previous contributions have addressed the use and detection of weak IVs by proposing alternative tests for instrument relevance. The issue of assessing instrument relevance is addressed by Hall et al. (1996). They mention that it is particularly an issue in applied work which tends to focus on instrument exogeneity rather than instrument relevance, as this is often hard to justify. They suggest to use a statistic to clarify a way to identify instrument relevance and weak instruments, but eventually remark that the use of t-statistics would only further distort inference.

Andrews et al. (2006) consider inference with weak instruments in the case of one endogenous variable and extend the theory to the cases of non-normal errors and multiple endogenous variables. In their work, they discuss and give an overview of several tests and confidence intervals which are claimed to be robust to weak instruments. They mention two approaches to inference with weak instruments: (i) using a combination of reporting estimates and standard errors with either pre-testing for weak instruments, or bias-corrected estimators and/or standard errors; (ii) reporting both estimates and fully robust confidence intervals using inverted robustness tests to test confidence intervals, where the authors recommend the second approach. Reviewing several hypothesis tests and confidence intervals which are claimed to be robust to weak IV estimation, the authors find that the conditional likelihood ratio (CLR) test is more reliable as compared to standard t-tests, but not, however, robust to non-normal errors.

Stock and Yogo (2005) address this distortion from weak inference by suggesting new critical values for the first stage F-statistic that account for possible biases. The usual construction of confidence intervals for the first-stage t-statistic $t \sim N(\mu, \sigma)$, where μ is the hypothesized value of the parameter, as $\pm 1.96 * \widehat{sd}$, for an assumed minimum value of $E[F]$ (the expectation of the first-stage F-statistic), results in a 90% confidence level, instead of the expected 95%, as a result of the distortion generated by instrument (*quasi*) irrelevance. Stock and Yogo (2005) further prove that, in order to obtain a 95% confidence interval using the common normally-distributed

t-statistic procedures, $E[F] > 142.6$ is a necessary condition.

Broadening the scope of these contributions, and as a part of their survey, Andrews et al. (2019) consider how practitioners are managing and detecting weak instruments in the recent empirical literature. They do this by examining a sample of publications in the American Economic Review from 2014-2018 that use the word "instrument" in their abstract. They find that weak instruments are indeed a problem in correctly estimating causal effects, and recommend that models containing a single endogenous regressor should base their instrument strength on the F-statistic. Furthermore, they recommend that in the case of a unique instrument, researchers should report AR confidence intervals.

Lee et al. (2021) further build on this by introducing the *tF-statistic*. The problem they identify is that many practitioners are still basing their results on the infamous rule-of-thumb threshold of 10 for the first-stage F-statistic, despite growing literature showing that this can lead to distortions in inference. Their proposal advances an already-existing literature on robust confidence sets for finite-sample biased IVs. Namely, they build on Staiger and Stock (1997) and Stock and Yogo (2005) to assess the methodological concerns in standard confidence intervals when strictly relying on pre-testing for the first-stage F-statistic. Focusing on *just-identified* models, they provide a "smooth" [sic] adjustment for t-ratio inference, employing the first-stage F-statistic. This contrasts with the common practice of employing *standard* t-ratio based inference after performing the $F > 10$ rule-of-thumb pre-check for instrument weakness and with Hall et al. (1996) who focused on a fixed statistic in assessing instrument relevance.

More specifically, they advance the so-called tF-statistic procedure to adjust confidence intervals at the 95% and 99% levels. They compare their proposal to the one proposed in Anderson and Rubin (1949), and find comparable minimum requirements as well as outcomes with three important enhancements. Firstly, finite expected-value of the confidence sets for $F > 3.84$; secondly, inherent robustness to heteroskedasticity and clustered-designs; and finally, no need to access the original data so long as the first-stage F-statistic is reported (as is common).

In sum, there is an increasing body of work that addresses the issue of weak instruments and their effect on accurate inference in IV settings, especially given the rise in popularity of IV estimation in applied economics fields. Most notably for us, Lee et al. (2021) have advanced a new approach which enables us to re-evaluate empirical contributions with minimal data requirements. Section 3 concerns this particular endeavour.

3 Data

The available literature on IV standard-error and confidence-interval correction is largely aimed at improving the theoretical foundations for future empirical applications, in particular allowing for more robust inference. Provided the centrality of policy evaluation, and the increasing popularity of RCTs as encouragement designs with imperfect compliance and natural experiments to leverage random or quasi-random variation in treatment assignment, IV estimation is extensively used. IV estimations are moreover particularly likely to qualify for the application of the tF critical value function adjustment, as they tend to be just-identified specifications, in which the instrument is random assignment to treatment or is, for instance, an indicator for a given side of the cutoff of the forcing variables in a regression discontinuity design. Importantly, however, this theory also allows for a retrospective assessment of published results and inference, which we now introduce.

3.1 Scope of re-evaluation

Our primary goal is to collect a sample of publications from which we can analyze the magnitude of necessary adjustment factors according to the tF procedure. In addition to the minimal fields necessary for the corrective procedure, we supplement our data scraping with regression specification and publication characteristics. This will allow us to not only analyze tF adjustment magnitudes but also perform broader correlation analyses and test which characteristics are significant predictors of said adjustment factors.

It is important to bear in mind throughout that, while this exercise leads to different standard errors used for inference, it is not intended to directly comment on the conclusions from these publications. Indeed though we initially tried to identify which specifications could be considered "main" specifications, or central to the thesis of the paper, this exercise proved altogether too subjective. As Lee et al. (2021), caution, it is "beyond the scope of [this research] to suggest whether any of the overall conclusions of the studies [...] would be altered in light of these adjustments." Thus, like them, we focus rather on "gauging to what extent the tF critical values are likely to impact the length of confidence intervals in research going forward."

3.2 Data description

The data collection process was designed to capture a representative sample of recent research in applied economics. To that end, we scraped contemporary publications published in one of the preeminent set of scholarly journals in economics, the *American Economic Journals (AEJ)*. Specifically, we collected single endogenous regressors, just-identified IV specifications from relevant publications subject to this ex-post revision.[1]

In our custom dataset (available here), each observation corresponds to a regression specification in a given paper. Importantly, 92.65% of our specifications report the first-stage F-statistic or provide alternate first-stage results sufficient to retrieve first-stage F as $t^2 \sim F$.

As shown in **Table 1**, we have aggregated specifications predominantly from *AEJ: Applied Economics*, with the remaining 26% and 18% coming from *AEJ: Economic Policy* and *AEJ: Macroeconomics*, respectively. Beginning in 2021, we scraped specifications from over the past decade, dating back to 2010. Further, in addition to the majority of specifications which employ standard IV estimation, we have also selected close to 16% employing IV within *Regression Discontinuity Design (RDD)* and *Regression Kink (RK)* frameworks, accounting for the boom in quasi-experimental methodological settings used for evaluation. Finally, our sampled papers cover a spectrum of topics which we formalized as one-to-many topic tags for each publication in our dataset. In **Table**

[1] While we found specifications with multiple instruments, the tF procedure was developed specifically for single endogenous variable, single instrument specifications which, *anecdotally*, were indeed the most prominent.

1 we present one aggregate topic per paper corresponding to a set of eight common categories, matched from their original topic tags.

Table 1: Per-Paper Descriptive Statistics

Variable	*Category*	*Count*	*Share*
Journal	*Applied Economics*	21	0.56
	Economic Policy	10	0.26
	Macroeconomics	7	0.18
Year	*2010*	2	0.05
	2011	2	0.05
	2012	2	0.05
	2013	1	0.03
	2014	1	0.03
	2015	3	0.08
	2016	3	0.08
	2017	6	0.16
	2018	8	0.21
	2019	5	0.13
	2020	3	0.08
	2021	2	0.05
Strategy	*IV*	32	0.84
	RK	3	0.08
	RDD	3	0.08
Topic	*Health*	3	0.08
	Education	9	0.24
	Labor	11	0.29
	Politics	3	0.08
	Tax	4	0.11
	Micro	2	0.05
	Law	1	0.02
	Macro	5	0.13

Note: N= 38. Unweighted distributions of categorical variables at the paper level. RK = Regression Kink; RDD = Regression Discontinuity Design. Source: own elaboration using our database.

Additionally, we have annotated author specific features such as the number of co-authors, the number of female co-authors, and the number of co-authors belonging to a top-10 institution (according to the Shanghai Ranking). We can observe the descriptive statistics of the relevant (non-categorical) variables in **Table 2**. We see that publications have two authors on average, and one out of four co-authors tends to be female as well as affiliated with a "top university." We can also observe that the range of the first-stage F-statistics and second-stage coefficients and standard deviations are large.

Table 2: Descriptive Statistics II

Variable	*Min*	*25th*	*Median*	*Mean*	*75th*	*Max*	*Std. Deviation*
Authors	1.00	2.00	2.00	2.08	3.00	4.00	0.78
Female authors	0.00	0.00	0.00	0.53	1.00	3.00	0.75
Top university	0.00	0.00	0.00	0.14	0.00	1.00	0.34
First stage F-stat	0.00	15.84	69.60	2532.64	260.00	308025.00	24089.09
IV Beta	-7622	-0.13	0.05	-182.99	0.50	102.57	1148.27
IV Std. Dev	0.00	0.04	0.11	23.36	0.40	919.00	132.59
Specifications	2.00	5.00	8.00	11.50	12.00	72.00	12.69

Note: N= 437. These statistics are weighted by the inverse of the number of specifications of each publication. Source: own elaboration using our database.

Finally, from **Table 2** we can see that there is considerable variation in the number of specifications per publication, ranging from 2 to 72. This showcases a relevant consideration regarding our dataset: that of heterogeneity, even within a single publication, in the presentation of regression results. While some papers present dozens of specifications, for example, others rely on the strictly necessary; moreover, in some papers, first-stages are only partially reported, if reported at all. Though some publications which belong to the latter category do present further data in their respective online appendices, the very different styles of presentation and degrees of completeness is notable. As a result, our data likely exhibits authors' sample selection as the number of specifications, which vary greatly, likely correlate with the aim of the paper, the hypotheses presented, and the significance of the findings. As we would expect this to influence our results, we will weight specifications proportional to the inverse of the number of specifications per publication to ensure our analyses are robust to few publications with outlying number of specifications.

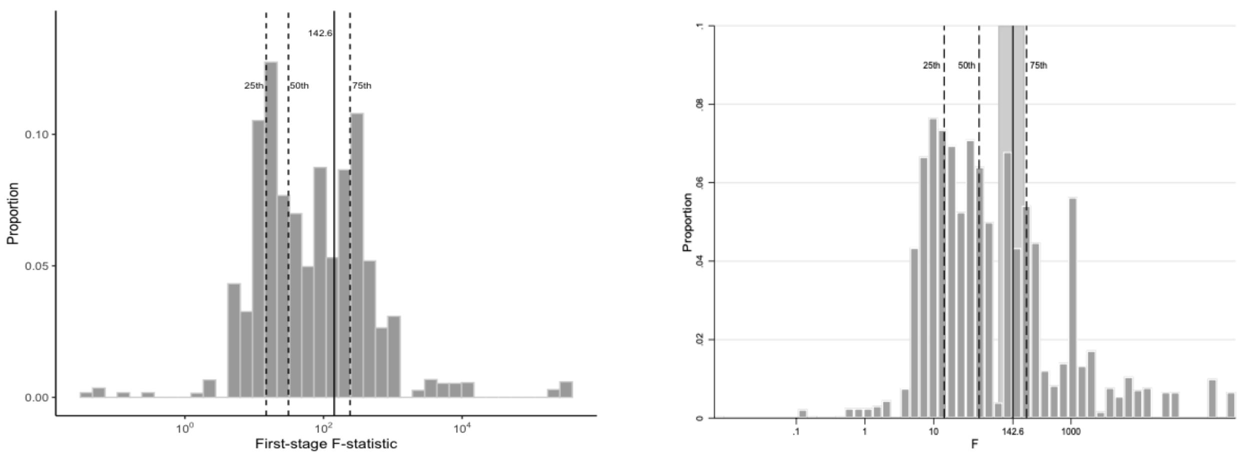
3.3 Data analysis

By construction, our dataset has all the relevant fields to apply the standard error correction proposed by Lee et al. (2021) using the tF critical value function as well as several additional fields for auxiliary analyses.

From all publications, we have aggregated a total of 468 specifications, 437 of which meet our desired criteria and report all necessary fields. From these 437 specifications, 262 (or 55.98%) are statistically significant at the 5%-level and 187 (or 42.79%) are significant at the 1%-level, using the reported standard errors.

Figure 1: Distribution of First-Stage F-statistics

(a) Our Sample **(b)** Lee et al. (2021)



Note: $N = 437$ in panel (a). Each observation corresponds to a specification using just-identified IV estimation in a paper published in AEJ journals between 2010 and 2020. Dashed lines correspond to the 25th (14.88), 50th (31.19), and 75th (241.77) percentiles of the distribution; solid line at 142.6. Observations are weighted by the inverse of the number of specifications per paper.
Source: own elaboration using our database; and Lee et al. (2021).

Figure 1 provides an overview of the distribution of F-statistics in our sample, as compared to Lee et al. (2021)'s original data. With over 75% of specifications having a first-stage F-statistic surpassing the infamous $F > 10$ rule-of-thumb, this presents partial evidence for the well-documented publication significance bias; however, we will not comment further on this topic as it is well outside the scope of our analysis.

Table 3: t^2 and First-Stage F-statistic

	$F \leq 10$	$F > 10$	Total
	21	241	262
$t^2 \geq 1.96^2$	(0.05)	(0.55)	(0.60)
	[0.07]	[0.64]	[0.71]
	27	148	175
$t^2 < 1.96^2$	(0.06)	(0.34)	(0.40)
	[0.05]	[0.24]	[0.29]
	48	389	437
Total	(0.11)	(0.89)	(1)
	[0.12]	[0.88]	[1]

Note: Counts and shares (unweighted in parentheses, weighted in square brackets) of specifications as a function of their significance using common-practice rules. Source: own elaboration using our database.

From **Table 3**, we can see that 48 specifications were using weak instruments at their baseline when looking at the first stage F-statistic. Considering our results using the tF-adjustment (see Table 4), we see that the number of specifications using weak instruments increases compared to the baseline.

In addition to reproducing the results in Lee et al. (2021), we decided to expand this analysis by examining correlations between auxiliary specification and paper characteristics and the magnitude of the associated tF adjustment factor. In particular, we have tested different subsets of variables from the following regression:

$$adj_{ij}^{\alpha} = \beta_0 + \beta_1 author_j + \delta_1 female_j + \beta_2 top\_uni + \beta_3 IV\_tstat^2 + \beta_4 n\_specs_j + \beta_5 fs\_fstat_{ij} + \sum_{m=1}^{3} \beta_m strategy_m + \sum_{k=1}^{10} \beta_k topic_j + \epsilon_{ij} \quad (2)$$

where $author$ indicates the number of authors per publication; $female$ indicates a dummy which equals 1 if at least one of the co-authors of a publication is female; top_uni is a dummy variable equalling 1 when at least one co-author is part of a university ranked in the Top 10 in the Shanghai Ranking; IV_tstat indicates the t-ratio of the second stage, for which we take the square so that negative values do not skew our results; n_specs indicates the number of specifications per publication; fs_fstat indicates the first stage F-statistic; $strategy$ indicates the IV design of the publication; $topic$ is a factor variable indicating the (sub)field of the paper. We have furthermore extended the regression by using time fixed-effects and time specific dummies. Subscripts i and j indicate specification and publication, respectively. α indicates the significance levels, meaning we will run the regressions for adjustment factors at both the 5%- and 1%-significance levels.

Running these regressions, we will compare the differences between the magnitude of adjustment factors on the 5%- and 1%-significance levels and their relation to the paper characteristics. We expect that a larger first-stage F-statistic will be negatively associated with the magnitude of the adjustment factor, as the adjustment factor is directly dependent on the F-statistic.

4 Results and Limitations

4.1 Results

We have computed the adjustment factor for the standard errors following the procedure proposed by Lee et al. (2021) using an iterative function, which assigns to each F-statistic the corresponding adjustment as retrieved from Table 3.a (respectively 3.b) in their paper for the 5%-level (respectively, 1%-level) adjustment. Notably, we employ a linear interpolation whenever the F-statistics falls within a given interval. The adjusted standard errors are then simply the product of the given standard error and the adjustment factor.

Table 4: Distribution of Standard Errors, tF-Adjustment Factors and Confidence Intervals

Variable	*Min*	*25th*	*Median*	*Mean*	*75th*	*Max*	*Std. Deviation*
tF-Adj. std. error (5%)	0	0.04	0.14	23.57	0.66	919.00	132.57
tF-Adj. std. error (1%)	0	0.05	0.19	25.57	1.52	973.22	140.36
Adjustment (5%)	1	1.00	1.05	1.51	1.43	9.52	1.41
Adjustment (1%)	1.06	1.06	1.25	3.78	2.17	35.37	8.27
Non-sign. (5%)	0	0.00	0.00	0.29	1.00	1.00	0.46
Non-sign. (1%)	0	0.00	0.00	0.44	1.00	1.00	0.5
Non-sign. (adj. 5%)	0	0.00	0.00	0.42	1.00	1.00	0.5
Non-sign.(adj. 1%)	0	0.00	1.00	0.69	1.00	1.00	0.47
CI length (5%)	0	0.15	0.42	91.56	1.55	3602.48	519.76
CI length (1%)	0	0.20	0.55	120.33	2.04	4734.69	683.11
CI length (adj. 5%)	0	0.17	0.55	92.39	2.60	3602.48	519.69
CI length (adj. 1%)	0	0.26	0.96	131.73	7.85	5014.03	723.14

Note: N= 437. These statistics are weighted by the inverse of the number of specifications of each publication. Source: own elaboration using our database.

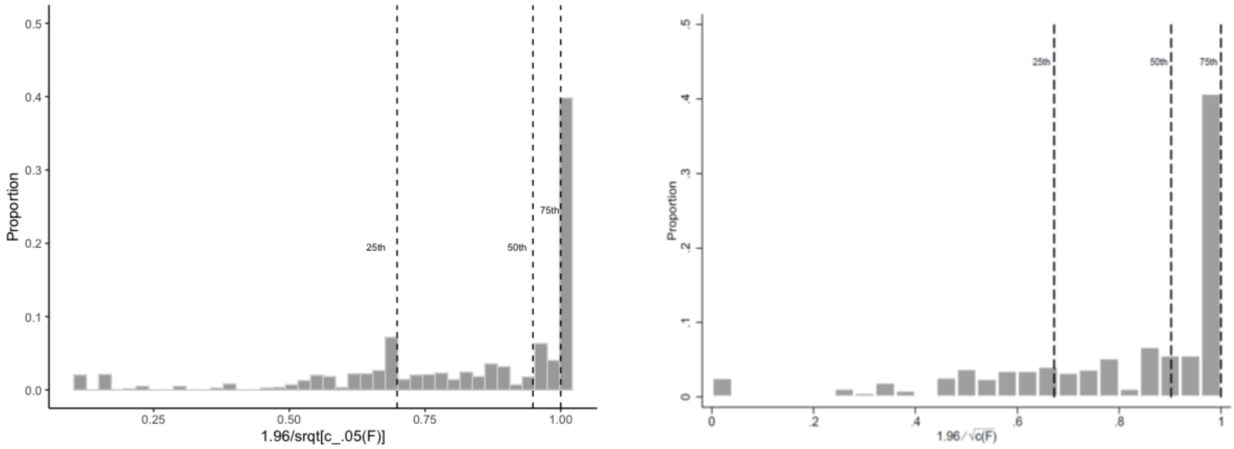
We then proceeded to calculate new confidence intervals with the tF standard errors. Using the adjusted standard errors, 197 of the 262 specifications that were originally significant at the 5%-level, remained significant. In other words, we found that nearly 25% of surveyed specifications dropped out of the 5% significance level when using the corrected tF-standard errors. Similarly, 101 of the 187 specifications that were originally significant at the 1%-level remained so, meaning nearly 30% dropped out. This is similar to the results in Lee et al. (2021), who find that about one-fourth of the initial statistically significant specifications were statistically insignificant when applying the adjustment factor. Hence, we see that even with our extension of the sample, the tF adjustment factor seems to have a substantive impact on statistical inference in applied work.

In **Figure 2**, we can see the distribution of the inverse of the adjustment factors at the 5% confidence level (naturally bounded in the support $(0, 1]$. Almost 40% of the specifications (weighting as aforementioned) report standard errors robust to weak instruments and finite-sample biases. Nonetheless, the rest of the sample of papers has a correction greater than 1, meaning that the consideration of the (ir)relevance of the instrument can have a significant impact in the inference framework. The average adjustment factor is 1.60 (1.52 when weighting by the inverse of the number of specifications at the paper level), which implies that inference in our sample suffers from weak instruments on average, and some of the results may be led by finite sample biases.

In **Figure 3** below we can observe the distribution of the inverse of the adjustment factors, similar to **Figure 2** but at the 1%-confidence level. We see that about 30% of the specifications report robust standard errors.

Figure 2: Distribution of tF-Adjustment Factors for the 5% Significance Level

(a) Our Sample **(b)** Lee et al. (2021)



Note: N = 437 in panel (a). Each observation corresponds to a specification. Dashed lines correspond to the 25th (0.69), 50th (0.95), and 75th (1) percentiles of the distribution. The x-axis measures the reciprocal of the adjustment factor for the sake of visualization. Observations are weighted by the inverse of the number of specifications per paper.
Source: own elaboration using our database; and Lee et al. (2021).

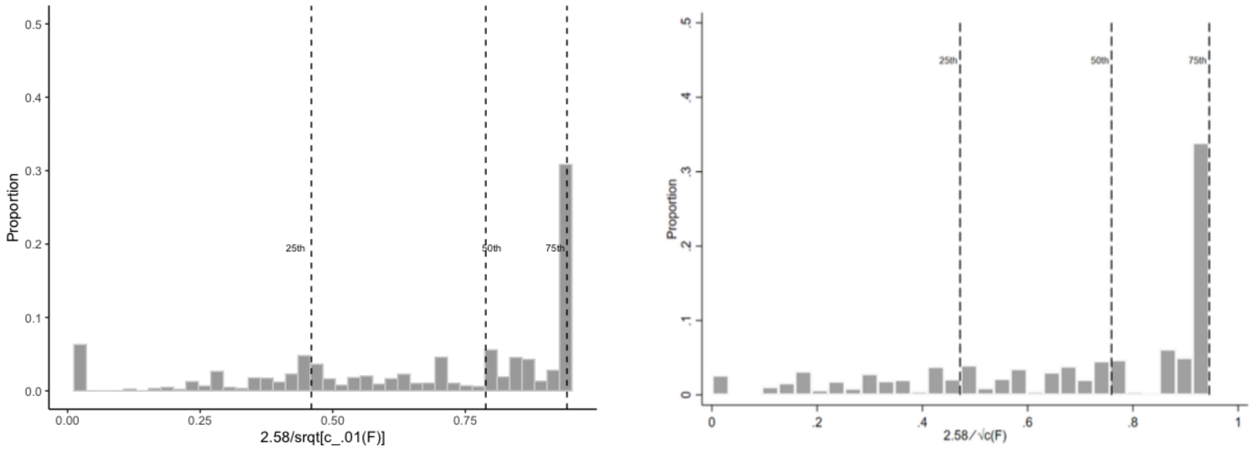
Here, the unweighted average adjustment factor is 3.98 (the weighted average adjustment factor being 3.78). Hence, we find even stronger adjustment factors at the 1%-level, further supporting the result that our sample is suffering from weak instruments on average.

In **Table 4**, we show the distribution statistics of the initial- and tF-adjusted standard errors. As previously stated, the average size of the adjustment factor is 1.52 at the 5% significance level, and 3.78 at the 1% significance level. Moreover, the maximum values of the adjustment factors are 9.52 and 35.37 at the 5%- and 1%-level, respectively, which are the maximum adjustment factors given by Lee et al. (2021). We can furthermore see that, at the 5% significance level, initially 29% of the specifications are not statistically significant and 44% of the specifications are significant at the 1%-level. When applying the tF-adjustment, we see that the share of non significance at the 5%-level and 1%-level increase to 42% and 69%, respectively, meaning that we see a 13- and 25 percentage point increase in non-significance. When comparing the length of the confidence intervals, we see that both on the 1%- and 5%-confidence levels the length of the confidence intervals increase. These results, again, imply that our extension of the work by Lee et al. (2021) supports the claim that the tF-statistic could have significant impact on statistical inference in future work.

Most importantly, we have replicated Figure 6 from Lee et al. (2021) in **Figure 4**. This graph captures the very essence of this endeavor: it classifies specifications as a function of their significance. Notably, it plots them in a color-coded fashion, where black points are not significant at the 5% level using the common-practice thresholds ($F > 10$ and $|t| > 1.96$), blue ones are so but become insignificant at the 5% level once the tF adjustment is applied, purple ones are significant at the 5% level after the adjustment, but the null of zero effect is rejected at the 1% with the adjustment, and finally, red ones are robustly significant at all levels, even after adjustments. The value added of the graph lies on it adjusting the axis to ensure visualization is bounded, and on combining the scatterplot of weighted specifications with the so-called *tF* critical value functions (Lee et al., 2021), which depict the minimum t^2 needed to make sure that the adjustment as a function of the first-stage F-statistic will not erase significance.

Figure 3: Distribution of tF-Adjustment Factors for the 1% Significance Level

(a) Our Sample **(b)** Lee et al. (2021)



Note: N = 437 in panel (a). Each observation corresponds to a specification. Dashed lines correspond to the 25th (0.46), 50th (0.79), and 75th (0.94) percentiles of the distribution. The x-axis measures the reciprocal of the adjustment factor for the sake of visualization. Observations are weighted by the inverse of the number of specifications per paper.
Source: own elaboration using our database; and Lee et al. (2021).

4.2 Extension: Regression Results

We have run regressions taking **expression 2** as a baseline. Results can be found in **Table 5** and **Table 6** (see **Appendix**). Since the identification strategy is not robust, these findings should be interpreted as correlational evidence: conditional on other observables, does any paper or specification trait statistically predict the magnitude of the *tF* adjustment factor? Put otherwise, are IV uses with certain characteristics more likely to be using an incorrect inference framework, without implying causation?

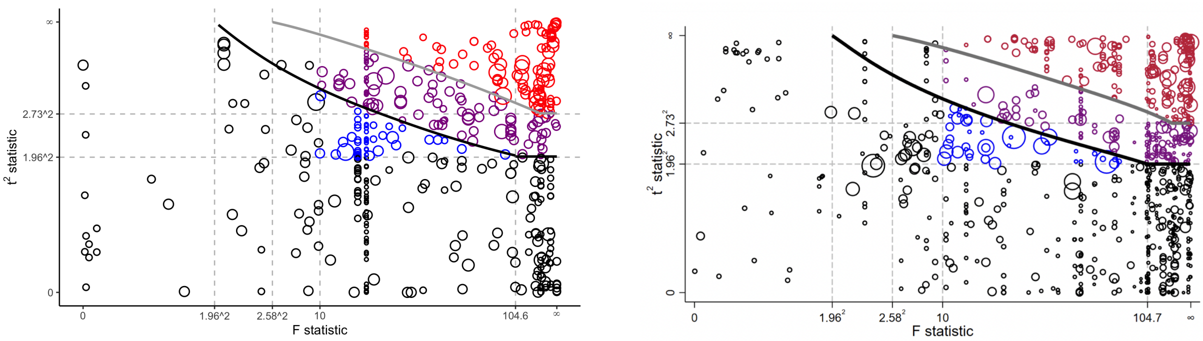
Firstly, in both regressions on the adjustment factors for 5%- and 1% significance levels, the first stage F-statistic does not statistically predict the adjustment factors. This is unexpected given that the *tF*-adjustment factors are a direct function of the F-statistic. However, conditional on the rest of covariates, all the variation may have been captured in other coefficients.

The number of authors per publication is positively associated with the adjustment factor on the 1%- and 5%-confidence levels, however only statistically significant at the 10%-level when looking at the 1% adjustment factors.

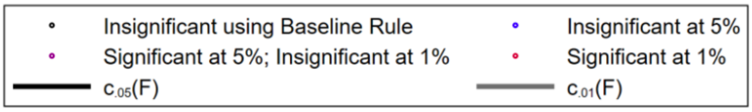
Furthermore, regression discontinuity designs seem to be negatively correlated with the adjustment factor at the 5% level, also showing a relatively high coefficient of -7.75 when looking at the 1%-adjustment factor. This may indicate that RD designs seem to be associated with a lower adjustment factor compared to standard IV designs, and may employ more robust instruments. We find the same results for the 1%-adjustment factor, however, when including the time trend and excluding year dummies this result is insignificant. We can furthermore see that having at least one female co-author in a publication is negatively associated with the magnitude of the 5%-adjustment factor, however, this is only significant at the 5%-level when including year dummies and is insignificant when looking at the 1%-adjustment factors. Having at least one co-author from a top-10 university seems to be associated to a 4 point lower 1%-adjustment factor when we include year dummies, at the 1%-confidence level. However, we should note that, though they are showing negative coefficients, none of the other regressions yield any significant results on a top-10 co-author being part of a publication. Moreover, we find that particularly topics in the fields of macro, politics and tax are positively associated with the size of the 5%-adjustment factor, as well as law when year dummies are included, again statistically significant at the 1%-level (excluding topics in the field of tax when no year dummies are included, which are significant only at the 5%- and 10%-levels). Topics in the field of Health only seem to be statistically significant at the 10%-level when including year dummies. When looking at the time trends of the tF-adjustment factor, we find that there is a negative association between the size of the adjustment factor over time, which is statistically significant at the 1%-level for all regressions. This indicates that publications in later years may be associated with a lower adjustment factor, and might point to an improvement of instruments used in statistical inference over time.

Figure 4: Comparing statistical significance using $c_{.05}(F)$ and $c_{.01}(F)$

(a) Our Sample **(b)** Lee et al. (2021) figure 6



Note: $N = 437$ *in panel (a). Observations are weighted by the inverse of the number of specifications per paper, indicated by the size of the circle. The vertical scale is* $\frac{\frac{t^2}{1.96^2}}{1+\frac{t^2}{1.96^2}}$ *and the horizontal scale is* $\frac{\frac{F}{10}}{1+\frac{F}{10}}$.
Source: own elaboration using our database; and Lee et al. (2021).



We have to be careful in interpreting these results, however, because of several factors. Firstly, assignment of topics is likely subjective given that these categories are generalized, so papers in a given field might still differ largely in topics that they cover. Secondly, we want to emphasize that these results should not be taken as final results; it should at most give an indication of possibilities for further analyses.

5 Conclusion

Taking into consideration that weak instruments generate inconsistencies for causal inference, we adopt the strategy put forward by Lee et al. (2021) and correct standard errors using a smooth function of the first-stage F-statistic which has desirable properties as compared to other confidence sets. Notably, we apply this correction to a sample of AEJ papers published between 2010-2021, and we document how approximately one quarter (respectively 30%) of the previously-significant specifications at the 5%-level (respectively 1%-level) cease to be so once we apply the tF adjustments. This gives further evidence to the notion that many contemporary practitioners continue to use an insufficient pre-test for instrument weakness which leads to biased inference.

6 Appendix

Table 5: Regression of the tF-Adjustment Factor at the 5% Significance Level on Paper and Specification Covariates

	Dependent variable:			
	tF-Adjustment Factor at the 5% Significance Level			
	(1)	(2)	(3)	(4)
Second Stage t^2	−0.0002 (0.0004)	−0.0004 (0.0004)	−0.0001 (0.001)	−0.0001 (0.001)
First Stage F-statistic	−0.0000 (0.0000)	−0.0000 (0.0000)	−0.0000 (0.0000)	−0.0000 (0.0000)
N. of Authors	−0.03 (0.14)	−0.02 (0.13)	0.29^{**} (0.15)	0.29^{**} (0.15)
$\mathbb{1}$(N. of Female Authors > 0)	−0.11 (0.16)	−0.11 (0.16)	0.10 (0.19)	0.10 (0.19)
N. of Authors from Top Uni.	−0.05 (0.21)	−0.27 (0.21)	-0.65^{***} (0.25)	-0.65^{***} (0.25)
Specifications	0.03^{***} (0.01)	0.02^{**} (0.01)	0.03^{**} (0.01)	0.03^{**} (0.01)
- Strategy (excl. normal IV)				
Regression Discontinuity	-1.36^{***} (0.34)	-0.72^{**} (0.36)	-1.42^{***} (0.37)	-1.42^{***} (0.37)
Regression Kink	-0.60^{*} (0.36)	−0.53 (0.35)	0.21 (0.43)	0.21 (0.43)
- Topic/Field (excl. education)				
Health	0.22 (0.29)	0.31 (0.28)	0.62^{*} (0.35)	0.62^{*} (0.35)
Labor	0.12 (0.19)	−0.03 (0.19)	0.15 (0.27)	0.15 (0.27)
Law	0.07 (0.46)	−0.12 (0.45)	1.53^{***} (0.57)	1.53^{***} (0.57)
Macro	1.28^{***} (0.24)	1.34^{***} (0.24)	1.81^{***} (0.27)	1.81^{***} (0.27)
Micro	0.63^{*} (0.36)	−0.10 (0.38)	0.50 (0.40)	0.50 (0.40)
Politics	1.75^{***} (0.29)	1.28^{***} (0.30)	1.31^{***} (0.42)	1.31^{***} (0.42)
Tax	0.53^{*} (0.29)	0.66^{**} (0.28)	0.75^{***} (0.28)	0.75^{***} (0.28)
- Year (excl. 2010)				
2011			-2.14^{***} (0.53)	-1.91^{***} (0.50)
2012			-1.71^{***} (0.41)	-1.25^{***} (0.36)
2013			-2.16^{***} (0.53)	-1.48^{***} (0.47)
2014			-1.93^{***} (0.68)	−1.01 (0.66)
2015			−0.75 (0.46)	0.40 (0.33)
2016			-2.77^{***} (0.54)	-1.39^{***} (0.39)
2017			-1.33^{***} (0.48)	0.27 (0.32)
2018			-2.01^{***} (0.32)	−0.18 (0.34)
2019			-2.33^{***} (0.42)	−0.27 (0.30)
2020			-1.74^{***} (0.44)	0.55 (0.41)
2021			-2.52^{***} (0.56)	
- Linear Time Trend		-0.12^{***} (0.02)		-0.23^{***} (0.05)
Constant	0.98^{**} (0.39)	1.92^{***} (0.43)	1.98^{***} (0.56)	1.98^{***} (0.56)
Observations	437	437	437	437
R^2	0.18	0.22	0.33	0.33
Adjusted R^2	0.15	0.19	0.28	0.28
Residual Std. Error	0.38	0.37	0.35	0.35
F Statistic	6.12^{***}	7.48^{***}	7.65^{***}	7.65^{***}

Note: *p<0.1; **p<0.05; ***p<0.01. The four regressions are weighted by the inverse of the number of specifications of each publication. Categorical variables are highlighted in bold, with the respective dummies for each category and a note for the excluded one. Standard errors in parentheses. Source: own elaboration using our database.

Table 6: Regression of the tF-Adjustment Factor at the 1% Significance Level on Paper and Specification Covariates

	Dependent variable:			
	tF-Adjustment Factor at the 1% Significance Level			
	(1)	(2)	(3)	(4)
Second Stage t^2	-0.001 (0.003)	-0.002 (0.003)	0.0000 (0.004)	0.0000 (0.004)
First Stage F-statistic	-0.0000 (0.0000)	0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)
N. of Authors	-0.50 (0.80)	-0.45 (0.78)	1.57^{*} (0.86)	1.57^{*} (0.86)
$\mathbb{1}$(N. of Female Authors > 0)	-1.02 (0.96)	-1.00 (0.93)	0.41 (1.12)	0.41 (1.12)
N. of Authors from Top Uni.	-0.15 (1.26)	-1.54 (1.25)	-4.00^{***} (1.47)	-4.00^{***} (1.47)
Specifications	0.13^{***} (0.05)	0.07 (0.05)	0.15^{**} (0.06)	0.15^{**} (0.06)
- Strategy (excl. normal IV)				
Regression Discontinuity	-7.10^{***} (2.01)	-3.07 (2.11)	-7.75^{***} (2.14)	-7.75^{***} (2.14)
Regression Kink	-2.92 (2.13)	-2.48 (2.07)	2.02 (2.52)	2.02 (2.52)
- Topic/Field (excl. education)				
Health	0.26 (1.70)	0.83 (1.65)	1.36 (2.02)	1.36 (2.02)
Labor	0.44 (1.13)	-0.47 (1.11)	-0.66 (1.60)	-0.66 (1.60)
Law	-0.83 (2.68)	-2.02 (2.62)	9.53^{***} (3.32)	9.53^{***} (3.32)
Macro	7.63^{***} (1.43)	7.99^{***} (1.39)	10.45^{***} (1.55)	10.45^{***} (1.55)
Micro	3.28 (2.12)	-1.31 (2.25)	3.00 (2.36)	3.00 (2.36)
Politics	9.19^{***} (1.73)	6.27^{***} (1.77)	5.86^{**} (2.47)	5.86^{**} (2.47)
Tax	2.10 (1.71)	2.87^{*} (1.67)	2.56 (1.65)	2.56 (1.65)
- Year (excl. 2010)				
2011			-12.79^{***} (3.10)	-11.34^{***} (2.91)
2012			-7.50^{***} (2.40)	-4.59^{**} (2.13)
2013			-12.86^{***} (3.09)	-8.49^{***} (2.72)
2014			-10.38^{***} (4.01)	-4.56 (3.84)
2015			-2.93 (2.71)	4.34^{**} (1.96)
2016			-17.79^{***} (3.17)	-9.06^{***} (2.29)
2017			-8.67^{***} (2.78)	1.51 (1.89)
2018			-10.76^{***} (1.89)	0.87 (2.00)
2019			-12.22^{***} (2.45)	0.87 (1.75)
2020			-9.90^{***} (2.56)	4.65^{*} (2.41)
2021			-16.00^{***} (3.26)	
- Linear Time Trend		-0.74^{***} (0.14)		-1.45^{***} (0.30)
Constant	2.17 (2.32)	8.06^{***} (2.53)	7.44^{**} (3.26)	7.44^{**} (3.26)
Observations	437	437	437	437
R^2	0.18	0.22	0.33	0.33
Adjusted R^2	0.15	0.20	0.29	0.29
Residual Std. Error	2.25	2.19	2.05	2.05
F Statistic	6.01^{***}	7.61^{***}	7.89^{***}	7.89^{***}

Note: **p<0.1;* ***p<0.05;* ****p<0.01. The four regressions are weighted by the inverse of the number of specifications of each publication. Categorical variables are highlighted in bold, with the respective dummies for each category and a note for the excluded one. Standard errors in parentheses. Source: own elaboration using our database.*

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